BGP Tracer: Two-Phase Joint Traffic-Routing Attribution

Phase 1: Event-Traffic Correlation · Phase 2: Pearl-PNS Causal Attribution · Q1/Q2/Q3 Verdicts

Data Inputs

Clou… RIP…
H…
5-m…

RIPE Atlas probes
Active reachability
traceroute

Coarse Screen

STL Decomposition
Weekly + daily + hourly
seasonal-trend-loess
89.7% ASes rejected

Phase 1

4 Parallel Detectors
zoom
Traffic | Outage | Leak | Hijack
Suspension - Temporal Correlation
c in {real_incident, silent_alert, ...}
c in {real_incident, silent_alert, ...}

Shared Cache

Shared Cache
Structured payload + confidence
Address (a, e, t, k) for audit
Degraded-mode log

Phase 2

Pearl-PNS + HCAF
zoom
5 Causal Evidence Channels
Bayesian Synthesis
Q3: RtoT, TtoR, or abstain
HCAF never hallucinates alone

Output

Q1/Q2/Q3
Verdicts +
LLM report
(renderer)

Phase 1 Zoom: Deterministic Arbitration

Traffic Detector
STL residual
anomaly in Ta(t)

Outage Detector
Routing withdrawal
+ Atlas drop

Leak Detector
AS-path anomaly
prefix overlap

Hijack Detector
Origin conflict
vs ROA / RPKI

Suspension
Evidential Gating
c in {real_incident, silent_alert, ...}
c in {real_incident, silent_alert, ...}

All four detectors run in parallel. Traffic detector: 0/7 hijack victims (key negative discriminator).

Confidence threshold τ=0.34 balances precision vs recall on the 29-event ground-truth.

Phase 2 Zoom: Pearl-PNS Causal Attribution

MSBD Breakpoint
τ* abstention boundary

PDM: Do-Calculus
CSS = E[Y|do(X=1)] - E[Y|do(X=0)]

CSS Bootstrap

Transfer Entropy

MSBD Breakpoint

Bayesian Synthesis
HCAF score (0–1)
Q3: RtoT, TtoR, or abstain
(high confidence to auto-escalate)

CSS alone: 3/29 events (10.3%). Bayesian synthesis (all 5 channels): 29/29 events.

HCAF never hallucinates alone. LLM is a narrative renderer, not a verdict source.

Output and NOC Loop

Q2 Verdicts (Correlation)
c in {outage, leak, hijack, roa, normal}

Q2 Traffic Flag
f_Q2 in {0,1}
CSS magnitude

Q3 Causation
f_Q3 in {0,1,abstain}
RtoT or TtoR

Evidence Report
LR, CI, p-value, HCAF, confidence

LLM Narrative
(renderer only)
structured verdict

Confidence >= 0.85 to auto-escalate to NOC.
Confidence < 0.85 to flag for human review.

Top-1 Accuracy
89.7% (29 events, v17 ground-truth)

Alert Reduction
9.3:1 (130 adjudicated live alerts)

End-to-end Latency
< 1 min per event (16-core server)

Causal Accuracy
HCAF: 3/29 alone to 29/29 synthesised